

BOTANY HUB MULTAI

B.Sc. I BOTANY — MINOR PAPER II

DIVERSITY OF PLANTS

UNIT III — Bryophyta and Pteridophyta

DETAILED TEACHER-STYLE NOTES

Definitions • Complete explanations • Classification tables • Life cycles • Flowcharts • Diagrams • Examples • Economic importance • Exam points

Prepared strictly from the uploaded 2025–26 Minor Paper II syllabus.

UNIT III — BRYOPHYTA AND PTERIDOPHYTA

The syllabus requires Bryophyta and Marchantia, plus Pteridophyta and Lycopodium, including characteristics, classification, morphology/anatomy, reproduction, life cycle and economic importance. ■filecite■turn11file0■L59-L69■

A. BRYOPHYTA

Definition — Bryophytes: Small land plants with a dominant gametophyte generation and a multicellular sporophyte that remains nutritionally dependent on the gametophyte to varying degrees; they reproduce by spores and require water for fertilization.

General characteristics

- Usually occur in moist or shaded habitats.
- No true vascular tissue in the usual textbook sense.
- Gametophyte is conspicuous and photosynthetic.
- Sporophyte remains attached to the gametophyte.
- Male gametes are motile; external water is important for fertilization.

Thallus structure

Bryophytes may be thalloid, as in many liverworts, or leafy, as in mosses. A thallus is a flattened, dorsiventral plant body without true root, stem and leaf differentiation.

Classification by Chapman and Chapman — broad study outline

Group	General body plan	Example
Liverwort-type bryophytes	Thalloid or leafy; often dorsiventral	Marchantia
Hornwort-type bryophytes	Thalloid gametophyte with horn-like sporophyte	Anthoceros
Mosses	Leafy gametophyte and capsule-bearing sporophyte	Funaria

Marchantia — detailed account

Marchantia has a dorsiventral, dichotomously branched thallus. The upper surface bears air pores leading to air chambers, while the lower side bears rhizoids and scales. Vegetative propagation occurs by fragmentation and gemmae; sexual reproduction is oogamous with male and female sex organs borne on specialized stalked receptacles.

Alternation of generations in Marchantia



Economic importance of bryophytes

- Pioneer colonization and soil formation.
- Water retention in ecosystems.
- Peat formation by Sphagnum and related uses.
- Environmental education and ecological indicators in some contexts.

B. PTERIDOPHYTA

Definition — Pteridophytes: Seedless vascular plants that reproduce by spores and show a dominant independent sporophyte with true roots, stems and leaves.

General morphology

- Vascular tissue is present.
- Sporophyte is dominant and independent.

- Spores are produced in sporangia.
- Gametophyte is generally small and independent in many forms.
- Fertilization still commonly requires water because sperm are motile.

Classification — exam outline

Major traditional class	Representative example	General feature
Psilopsida	Psilotum	Primitive vascular organization
Lycopsida	Lycopodium	Microphyllous leaves
Sphenopsida	Equisetum	Jointed stem
Pteropsida	Ferns	Large fronds; sporangia commonly in sori

Lycopodium — morphology and anatomy

Lycopodium is a vascular cryptogam with a sporophytic plant body differentiated into roots, stem and small microphyllous leaves. Sporangia are associated with specialized sporophylls, often arranged into strobili in many species.

General life cycle of Lycopodium



Economic importance of pteridophytes

- Ornamental ferns are economically important.
- Some species are used in traditional practices and local products.
- They contribute to ecological succession and soil stabilization.
- Fossil pteridophytes are important for understanding plant evolution.

Exam focus: Unit III long answer: define the group → list general characters → explain plant body → reproduction → life cycle with ploidy → economic importance. The syllabus specifically names Marchantia and Lycopodium, so their diagrams and cycles are priority topics. ■filecite■turn11file0■L59-L69■